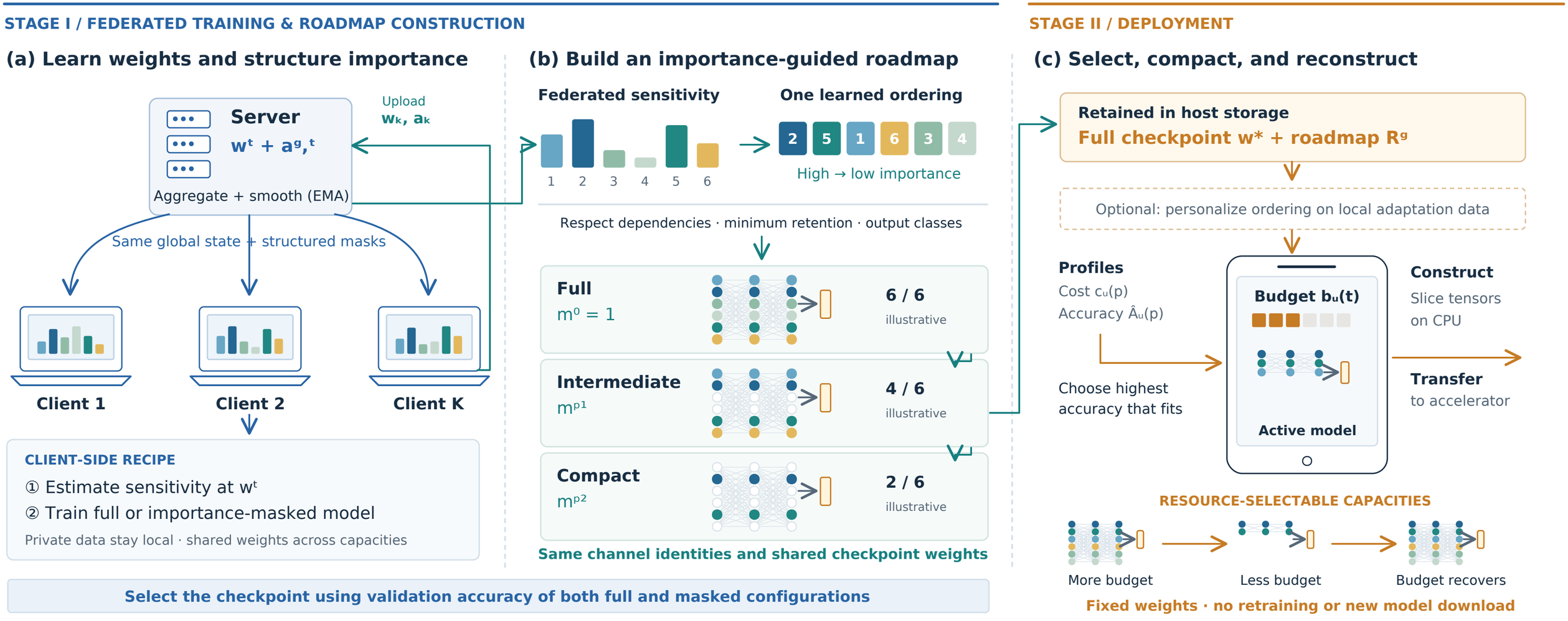

STAGE I / FEDERATED TRAINING & ROADMAP CONSTRUCTION
STAGE II / DEPLOYMENT
(a) Learn weights and structure importance
Server
w^t + a^g,^t
Aggregate + smooth (EMA)
Upload
w_k, a_k
Same global state + structured masks
Client 1
Client 2
Client K
CLIENT-SIDE RECIPE
① Estimate sensitivity at w^t
② Train full or importance-masked model
Private data stay local · shared weights across capacities
(b) Build an importance-guided roadmap
Federated sensitivity
One learned ordering
1 2 3 4 5 6
2 5 1 6 3 4
High → low importance
Respect dependencies · minimum retention · output classes
Full
m^0 = 1
6 / 6
illustrative
Intermediate
m^p1
4 / 6
illustrative
Compact
m^p2
2 / 6
illustrative
Same channel identities and shared checkpoint weights
Select the checkpoint using validation accuracy of both full and masked configurations
(c) Select, compact, and reconstruct
Retained in host storage
Full checkpoint w* + roadmap R^g
Optional: personalize ordering on local adaptation data
Profiles
Cost c_u(p)
Accuracy Â_u(p)
Choose highest accuracy that fits
Budget b_u(t)
Active model
Construct
Slice tensors on CPU
Transfer
to accelerator
RESOURCE-SELECTABLE CAPACITIES
More budget
Less budget
Budget recovers
Fixed weights · no retraining or new model download